Guillermo Albert García

Backend-first Full Stack Developer · Java / Spring / Angular

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Full Stack Developer with 2+ years of experience in international enterprise projects using Java, Spring and Angular. Beyond day-to-day work, I run a self-hosted infrastructure (Proxmox, 9 LXC containers) where an ETL pipeline against the Catapult API, an LLM-assisted invoice pipeline and an operations assistant built on LLM APIs for a real SME (self-hosted RAG, human-in-the-loop) run in production. I come from biology research (3 publications, 1 in a peer-reviewed journal). Native Spanish, bilingual French (French Baccalauréat), C1 English.

WORK EXPERIENCE

Full Stack Developer — CGI — Sep 2026 – Present
Banking sector (France) · Java / Angular

Full Stack Developer — Inetum — Mar 2024 – Sep 2026
Alicante, Spain (Hybrid) · International team (~10 people) · Internship → permanent contract

Two platforms for **FranceAgriMer** (French public agency under the Ministry of Agriculture) managing agricultural aid, subsidies and payments to farmers across France, moving millions of euros annually.

- **Legacy project (JSF / Java 8 / Oracle):** maintenance and evolution of a production application — refactoring of controllers and services, conditional JSF views, **JasperReports** templates for the official documents of the 2025/2026 campaigns, and **15+ corrective PL/SQL scripts** on a live database (migrations and unblocking case files).
- **New project (Angular / Spring / JPA / REST API):** built complete *end-to-end* flows through go-live — from data model to controller (entity, DTO, mapper, repository, service), complex business logic (algorithmic amount distribution, pre-settlement process), schema evolution with **Liquibase**, and Angular components and services with state management and real-time calculations.
- Technical-debt reduction with **SonarQube** (bugs, security vulnerabilities and code smells), coverage with **JUnit** (backend) and Angular component tests.

Science Teacher — Lycée Français d'Alicante — Sep 2023 – Sep 2024

- Taught science at upper-secondary level (French baccalauréat) in a **bilingual (French/Spanish) environment**, while starting at Inetum.

PERSONAL PROJECTS & INFRASTRUCTURE

AI-powered operations assistant for a guided-tours SME — Jun 2026 – Present
Business application in production for a real company, built on top of LLM APIs · Python/FastAPI, PostgreSQL + pgvector, React, Anthropic API

- **Self-hosted RAG** over 9,571 chunks with **fully local embeddings** (GDPR: business emails never leave the network), distilled from **13,047 emails**; incoming-email classification and reply drafting with context and example injection (few-shot).
- **Verifiable human-in-the-loop**: a single SMTP exit point, reachable only after human confirmation, with an audit record written before sending; critical validations are deterministic, kept out of the prompt. **8,000+ tests** and spec-driven development (13 specs) assisted by AI agents (Claude Code); multi-provider LLM client with two cost tiers built in.

MAMS — My Athlete Monitoring System — 2025 – Present
Athlete monitoring platform (AMS) for semi-professional clubs and academies · github.com/GuillermoAlbert/mams-architecture

- Backend-focused full stack architecture: **Spring Boot** (Java 21), **PostgreSQL**, authentication with **Spring Security + JWT**, multi-level permissions model and schema design with Liquibase. Custom design system and UI components (body map, player cards, dual colour scales) on a **React frontend (built with AI agents under my direction)**.
- Daily wellness and RPE check-ins with a coach dashboard and alerts (21 REST endpoints, per-team access control); data integrity enforced by the database itself (CHECK and UNIQUE constraints, generated columns).

Proxmox Home Server + Tailscale Network — 2024 – Present
Self-hosted infrastructure · 9 LXC containers in production/dev · Proxmox VE, LXC, Tailscale, Docker, Debian/Ubuntu · Python automations built with AI agents, supervised and operated by me · github.com/GuillermoAlbert/homelab

- **Automated invoice pipeline** (Python, IMAP, LLM, SQLite): downloads invoice PDFs from the mailbox over IMAP, extracts structured data via AI and validates it with deterministic arithmetic, routing low-confidence invoices to

review; Excel exports generated from the DB. 115-test pytest suite; diagnosed and fixed a SQLite WAL bug in production.

- **GPS ETL: Catapult → PostgreSQL + Power BI** (Python, PostgreSQL 16, REST API): centralises data from 6 teams with upserts, Flask admin panel for config and historical backfills, read-only access via Tailscale for client Power BI.
- **Monitoring dashboard**: self-built web panel with real-time ETL status (log parsing), action buttons and an alert bot. Proactively detected a stopped cron job in production.
- **Hands-on incident management**: broken Docker network after an unexpected host reboot, DNS false positives in AdGuard, NAT config with iptables. Encrypted backups to Google Drive with restic/rclone.

TECHNICAL SKILLS

Languages	Java, TypeScript, JavaScript, SQL, HTML, CSS
AI / Agents	Claude Code (skills, hooks), Claude API, MCP, Codex, Opencode
AI / LLM	LLM APIs (Anthropic Claude, multi-provider client), RAG (pgvector, local embeddings), prompt & context engineering, human-in-the-loop, evaluation against real data
AI-assisted development	Python (ETL / automation) and React — code generated with AI agents (Claude Code) under my direction: architecture, code review, testing and production operation are my own
Backend	Spring (MVC, Boot), Spring Security (JWT), JPA/Hibernate, JSF, REST APIs
Frontend	Angular, TypeScript, HTML5, CSS3
Databases	PostgreSQL 16 (+ pgvector), Oracle, SQLite
Infrastructure	Proxmox VE, Docker/Compose, LXC, Linux (Debian/Ubuntu), Tailscale
Tools	Git, GitLab, Liquibase, JUnit, SonarQube, JasperReports, Agile

RESEARCH

Biology research

2019 – 2022

CSIC — Doñana Biological Station · Luomus — Finnish Museum of Natural History (Univ. of Helsinki)

- Sequencing and analysis of biodiversity data. **3 publications**, including AI-based detection of tagged animals in camera-trap images (*Ibis*, 2022).

EDUCATION

Higher Technician in Web Application Development (DAW)

IES Mare Nostrum, Alicante · 2022–2024

MSc Secondary Education Teaching — Biology & Geology

Universidad de Alicante · 2020–2021

MSc Biodiversity: Conservation and Evolution

Universitat de València · 2021–2022

BSc Biology

High Academic Performance group, taught in English · Erasmus: Univ. Helsinki

Universidad de Alicante · 2016–2020

PUBLICATIONS

Santangeli, A., Chen, Y., Boorman, M., Sales Ligero, S. & **Albert García, G.** (2022). Semi-automated detection of tagged animals from camera trap images using artificial intelligence. *Ibis*, 164(4), 1123–1131. doi:10.1111/ibi.13099

Aznar-Avendaño, F.J., **Albert-García, G.**, Barón-Rodríguez, P., et al. (2022). Teaching Parasitology in Biology Degrees: From Subjects to Principles. *INTED2022 Proceedings*, pp. 2481–2487. doi:10.21125/inted.2022.0726

Albert, G. & Arcos, J. (2015). Hormigas del Parque Natural de Serra Gelada y citas interesantes para la mirmecofauna alicantina (Hymenoptera, Formicidae). *Iberomyrmex*, 7, 3–6.

LANGUAGES & OTHER

Spanish — Native · **French** — Bilingual (Baccalauréat)

English — C1 (degree taught in English) · Cambridge First certificate

JSF 2.2 – Build a DB App (Udemy, 2024)